

Aaron Daniel Kartmann

 LinkedIn

 GitHub

 Portfolio

Experimental Condensed Matter Physicist

 Nijmegen, The Netherlands

MSc Physics, specialization in Quantum Matter
Radboud University Nijmegen

RESEARCH PROFILE

Experimental physicist with an MSc in Physics, specializing in Quantum Matter, from Radboud University Nijmegen. My research experience includes electronic transport and torque magnetometry at high magnetic fields and cryogenic temperatures, as well as laser-based magneto-optical measurements on magnetic thin films. I am interested in experimental research on optics, quantum materials, and nanoscale devices, particularly where a fundamental understanding of physical properties can contribute to future technologies.

RESEARCH EXPERIENCE

Master's Research Project

Sep 2025 – Aug 2026

Correlated Electron Systems group, HFML-FELIX, Nijmegen

Grade: **9/10**

- Investigated the magnetoresistance, superconductivity, and quantum oscillations of quantum materials using electronic transport measurements and cantilever torque magnetometry at high magnetic fields and cryogenic temperatures.
- Compared the magnetoresistance and Fermi surface of pristine PtBi₂ with Pb-doped PtBi₂. Found evidence for 2D superconductivity in pristine PtBi₂.
- Showed that both compressive and tensile uniaxial strain can tune the magnetoresistance and Fermi surface of WTe₂.
- Investigated the magnetoresistance of MnSb₂Te₄. Found an unusual low-temperature upturn in the electrical resistance.
- Gained laboratory experience with superconducting and Bitter magnets, cryogenic systems, vacuum equipment, and sample preparation.
- Developed skills in experimental setup, measurement troubleshooting, data analysis, and interpretation of experimental results.

Supervisor: Dr. Steffen Wiedmann.

Bachelor's Research Project

Feb 2024 – Jul 2024

Spectroscopy of Solids and Interfaces group, Radboud University Nijmegen

Grade: **8.5/10**

- Demonstrated helicity-dependent all-optical switching (HD-AOS) of magnetization in Pt/Co/Pt thin films on a 50 nm-thick silicon nitride membrane, extending the established effect from conventional glass substrates to a novel thin membrane platform.
- Compared the laser fluence thresholds for HD-AOS of samples on the silicon nitride and glass substrates.
- Gained laboratory experience in optics and working with lasers, specifically Faraday microscopy and Faraday rotation measurements.

Supervisor: Prof. Theo Rasing.

EDUCATION

MSc Physics, specialization in Quantum Matter

2024 – 2026

Radboud University Nijmegen

Final grade: **8.8/10, Cum Laude.**

BSc Physics and Astronomy

2021 – 2024

Radboud University Nijmegen

Final grade: **8.3/10, Cum Laude.**

TECHNICAL SKILLS

Programming

Python (NumPy, data analysis, numerical simulations, machine learning); C++ (object-oriented programming, computationally efficient programming, Python integration)

Laboratory

Cryogenic systems (⁴He, ³He, liquid nitrogen), vacuum equipment, sample preparation, electrical contacting

Data analysis and software

OriginPro, LaTeX, Linux, Git, Microsoft Office, HTML, CSS

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Thermodynamics (Warmteleer)

2023/24 and 2024/25

Radboud University Nijmegen

- Led tutorials, guided first-year students through physics exercises, answered questions, and graded assignments.
- Developed experience in explaining physical concepts clearly.

LANGUAGES

German: C2

English: C1, Cambridge C1 Advanced (**Aug 2021**)

Dutch: C1 proficiency, RU-NT2-exam (B2 level) (**Jun 2021**) **French:** A1